

PRD – Task 01

Data Cleaning and Visualization

Project Requirements Document for completing the internship/hackathon-style data science task efficiently, correctly, and with a submission-ready result.

Task	01 – Data Cleaning and Visualization
Primary Goal	Clean an open-source dataset and perform meaningful EDA
Core Stack	Python • Pandas • NumPy • Matplotlib • Seaborn
Optional	Plotly / Power BI / Tableau for interactive dashboards
Expected Outcome	Clean dataset + EDA visuals + findings + reproducible project

Source requirement captured from the provided task brief: handle missing values, duplicates, incorrect data types and outliers; then perform EDA using histograms, bar charts, box plots and correlation heatmaps to discover trends and patterns.

1. Project Overview

The project is a complete data-cleaning and exploratory-data-analysis (EDA) workflow on one open-source dataset. The work must demonstrate that raw data can be transformed into a reliable analytical dataset and then communicated through clear visualizations.

Primary objective

- Select a relevant, open-source dataset with enough numeric and categorical fields to support meaningful EDA.
- Inspect the raw dataset before making changes.
- Clean missing values, duplicates, incorrect data types and extreme/outlier observations using justified methods.
- Generate required EDA plots and interpret what each plot reveals.
- Document decisions so another person can reproduce the analysis.

Problem statement

Raw datasets commonly contain incomplete records, duplicate rows, inconsistent types, invalid values and extreme observations. The task is to build a reliable cleaning pipeline and use visualization to identify patterns, relationships and anomalies without making unsupported conclusions.

2. Scope and Requirements

In scope

- Dataset acquisition from a legitimate open-source/public data source.
- Data loading and initial profiling.
- Missing-value analysis and treatment.
- Duplicate detection and removal.
- Data-type correction and consistency checks.
- Outlier identification and justified treatment.
- Univariate, categorical and relationship-based EDA.
- Required visualizations: histogram, bar chart, box plot and correlation heatmap.
- Optional interactive dashboard using Plotly, Power BI or Tableau.
- Final findings, limitations and reproducible project documentation.

Out of scope

- Building a production ML model is not required by the brief.
- Do not delete observations simply to make graphs look better.
- Do not claim causation from correlation or descriptive plots.
- Do not use an interactive dashboard as a substitute for the required core EDA.

3. Dataset Selection Requirements

Choose a dataset that makes the cleaning and visualization work visible. A very small or perfectly clean dataset may not demonstrate the required skills.

Criterion	Requirement / Guidance
Source	Open-source/public dataset with a clear source and usage terms.
Size	Prefer a dataset large enough to reveal patterns, but manageable on a student laptop.

Columns	Include both numeric and categorical variables where possible.
Quality issues	Prefer realistic missing values, duplicates, inconsistent types or outliers.
Domain	Choose a domain you can explain: healthcare, finance, retail, education, transport, social media, etc.
Documentation	Record source URL/name, dataset version/date if available, and original row/column count.

4. Functional Requirements

FR-01 — Load and inspect data

- Load the raw file with Pandas.
- Display shape, column names, sample rows and descriptive statistics.
- Check data types and identify numeric vs categorical columns.
- Create a baseline data-quality summary before cleaning.

FR-02 — Handle missing values

- Calculate missing count and percentage for every column.
- Choose drop, fill, or domain-based treatment only after assessing the impact.
- Document the reason for each treatment.
- Verify that the final dataset has no unintended missing values in required fields.

FR-03 — Remove duplicates

- Detect exact duplicate rows.
- Measure how many duplicates exist before removal.
- Remove duplicates where they represent repeated records.
- Re-check row count after the operation.

FR-04 — Fix data types and invalid values

- Convert dates to datetime when applicable.
- Convert numeric-looking strings to numeric types where appropriate.
- Standardize categorical text such as capitalization/spacing when needed.
- Check for impossible or invalid domain values and document the rule used.

FR-05 — Detect and address outliers

- Use box plots and/or an IQR-based method for numeric variables.
- Investigate extreme values before removing or capping them.
- Keep legitimate extreme observations when they are meaningful.
- Document every outlier treatment and its effect on row/value counts.

5. EDA and Visualization Requirements

The EDA should move from simple distributions to comparisons and relationships. Every important chart must have a descriptive title, readable axis labels and a short interpretation.

Visualization	Purpose	Minimum expectation
Histogram	Understand distribution, skewness and concentration	Use at least 2–3 important numeric variables.

Bar chart	Compare categories or group-level counts/values.	Use at least 2 meaningful categorical comparisons.
Box plot	Compare distributions and identify outliers.	Use important numeric fields, optionally grouped by category.
Correlation heatmap	Explore linear relationships among numeric variables	Include all relevant numeric features and interpret notable relationships.
Optional scatter/line plot	Explore pairwise or time-based patterns.	Use when the dataset contains suitable variables.

Visualization quality checklist

- Use readable font sizes and legends.
- Avoid unnecessary 3D charts and decorative effects.
- Use consistent units and meaningful labels.
- Do not overload one figure with too many categories.
- Include the key observation directly below or beside each important figure.

6. Recommended Technical Stack

- **Python:** primary programming language.
- **Pandas:** loading, cleaning, transformation and tabular analysis.
- **NumPy:** numerical operations and supporting calculations.
- **Matplotlib:** core plotting and figure customization.
- **Seaborn:** statistical visualization, especially box plots and heatmaps.
- **Optional Plotly:** interactive charts/dashboard.
- **Optional Power BI/Tableau:** interactive dashboard if you want to extend the submission.
- **Jupyter Notebook or VS Code:** recommended development environment.
- **Git/GitHub:** recommended for version control and submission evidence.

Suggested environment setup

```
python -m venv .venv
# Windows: .venv\Scripts\activate
pip install pandas numpy matplotlib seaborn jupyter
```

7. End-to-End Execution Workflow

Phase	Activity	What to do
Phase 1	Select dataset	Confirm source, domain, license/usage information, size and column suitability.
Phase 2	Raw-data audit	Load data; record shape, types, missingness, duplicates, unique counts and summary statistics.
Phase 3	Cleaning	Handle missing values, duplicates, types, invalid values and outliers with documented decisions.
Phase 4	Validation	Compare before/after row counts, null counts, dtypes, ranges and duplicate counts.
Phase 5	EDA	Generate distributions, category comparisons, outlier views and correlation analysis.
Phase 6	Insights	Write concise observations supported by the charts; separate findings from assumptions.
Phase 7	Optional dashboard	Build interactive views in Plotly/Power BI/Tableau if time permits.
Phase 8	Package	Clean notebook/script, cleaned dataset, figures, README and final report/presentation.

8. Data-Quality Checks

Before cleaning

- Record original row and column count.
- List missing values by column.
- Record duplicate-row count.
- Record original dtypes.
- Check numeric ranges and category values.

After cleaning

- Confirm expected row/column count and explain changes.
- Confirm intended missing-value treatment.
- Confirm duplicates were handled.
- Confirm corrected dtypes.
- Confirm outlier decisions and quantify their impact.
- Run a final describe()/info() review before EDA.

9. Expected Deliverables

Deliverable	Required content
01. Source dataset	Original dataset or documented source reference.
02. Cleaned dataset	CSV/Excel or equivalent cleaned data file, if redistribution is permitted.
03. Notebook/script	Reproducible Python workflow from loading through EDA.
04. EDA figures	Required plots exported or embedded in the notebook/report.
05. Findings	Concise, evidence-based observations and limitations.
06. README	Project purpose, dataset source, setup, execution steps and results summary.
07. Optional dashboard	Plotly, Power BI or Tableau dashboard and instructions.
08. Final submission	Organized folder/GitHub repository with no unnecessary files or secrets.

10. Suggested Project Structure

Use a simple structure so the evaluator can quickly understand and reproduce your work:

```
data-cleaning-visualization/
├── data/
│   ├── raw/
│   └── processed/
├── notebooks/
│   └── task_01_eda.ipynb
├── src/
│   ├── cleaning.py
│   └── visualization.py
├── outputs/
│   └── figures/
├── cleaned_dataset.csv
├── README.md
└── requirements.txt
```

Minimum recommended notebook sections

- 1. Project objective
- 2. Dataset source and description

- 3. Imports and configuration
- 4. Load raw data
- 5. Data overview/profile
- 6. Missing-value analysis
- 7. Duplicate analysis
- 8. Data-type and consistency fixes
- 9. Outlier analysis/treatment
- 10. Validation after cleaning
- 11. EDA visualizations
- 12. Key findings and limitations
- 13. Conclusion

11. Acceptance Criteria

- The project uses a documented open-source dataset.
- Raw data is inspected before cleaning.
- Missing values are measured and handled with a stated rationale.
- Duplicates are detected and addressed.
- Incorrect/inconsistent data types are corrected where necessary.
- Outliers are investigated and handled or retained with justification.
- At minimum, the submission contains histograms, bar charts, box plots and a correlation heatmap where the data supports them.
- Charts are readable, correctly labeled and accompanied by interpretations.
- The cleaned dataset and code can be reproduced by another person.
- The final README/report clearly states dataset source, methodology, findings and limitations.

12. Efficiency Plan / Suggested Timeline

Session	Focus	Output
1	Dataset selection + raw audit	Dataset source + initial profile
2	Missing values + duplicates	Cleaning logic and before/after metrics
3	Data types + invalid values + outliers	Validated cleaned dataset
4	EDA visualizations	Required plots + observations
5	Insights + optional dashboard	Findings and dashboard if time permits
6	README + final QA + submission	Complete, organized final project

13. Final Submission Checklist

- ■ Dataset source is documented.
- ■ Raw row/column count is recorded.
- ■ Missing-value analysis is included.
- ■ Duplicates are analyzed and handled.

- ■ Data types are checked and corrected.
- ■ Outliers are investigated with box plots/IQR or an appropriate method.
- ■ Histograms are included.
- ■ Bar charts are included.
- ■ Box plots are included.
- ■ Correlation heatmap is included.
- ■ Each important chart has an interpretation.
- ■ Cleaned data is saved where permitted.
- ■ Notebook/script runs from start to finish.
- ■ README explains setup and execution.
- ■ No passwords, API keys or unnecessary temporary files are committed.
- ■ Optional dashboard is clearly separated from the required core task.

Recommended final report structure

Introduction → Dataset → Data Quality Problems → Cleaning Methodology → EDA → Key Findings → Limitations
→ Conclusion → Tools/Technologies → Repository/Submission Details

14. Practical Guidance for Strong Results

- **Do not clean blindly:** inspect the data first and record why each transformation is needed.
- **Keep before/after evidence:** show counts for nulls, duplicates and rows before and after cleaning.
- **Make every chart answer a question:** e.g., “Which categories dominate?” or “Which variables move together?”
- **Interpret, don't just plot:** one or two sentences explaining the useful observation is more valuable than a chart alone.
- **Be transparent:** mention limitations, sampling issues, possible bias and any assumptions.
- **Prioritize the required task:** finish cleaning + required EDA first; build an interactive dashboard only after the core requirements are complete.
- **Reproducibility matters:** use clear code, consistent naming, a requirements file and a README.

Definition of done

The task is complete when a reviewer can open the project, understand the dataset source, see what was wrong with the raw data, verify how it was cleaned, inspect the required EDA visualizations, read the key findings, and reproduce the analysis without needing undocumented steps.